AutoSRE: cloud handoff report

Live validation on AWS and what the cloud role needs to do. Repo: vaishnavi-nss/autosre-devops.

Where the code is

- **ml-cloud-additions** (1bd539b): the PR branch. Adds files only; nothing on *main* is modified.
- **ml-and-cloud-validation** (da03baa): full history, including the files that overlap with *main*.
- The pull request is not opened yet; base branch must be *main*.

Live validation results

AWS Academy lab, one t3.large with minikube and kube-prometheus-stack. Faults were injected through the CHAOS_ENABLED endpoints; the alert rules, Alertmanager, webhook and engine were the real pipeline.

Measure	Result
Injected-fault trials (two runs)	32
Wrong actions by the rule engine	0
Time for an incident to reach the engine	about 2 minutes after injection
Known incidents a 0.75 gate would act on	about 40-46%
Alerts clear after a fault ends	about 5 minutes (rate()[5m] windows)

- The deciding alert fires first and alone, so the first webhook scores 0.55-0.60. Disk-full never reached 0.75 on the first webhook.
- Set the dispatcher cooldown to at least 10 minutes: alerts keep firing about 5 minutes after the fault ends.

What to do by hand

These files already exist on *main*, so the PR leaves them out:

- **scripts/build-images.sh**: also build autosre/second-opinion:dev.
- **scripts/deploy.sh**: wait for the second-opinion rollout.
- **k8s/correlation-engine.yaml**: set SECOND_OPINION_URL and SECOND_OPINION_MODE=shadow.
- **k8s/second-opinion.yaml** is new and included in the PR (256Mi request, 512Mi limit).

Fix that affects deployment

The engine Dockerfile did not copy second_opinion.py, so the engine pod crash-looped at startup. Fixed; tests/test_dockerfile.py now fails if any imported module is not copied into an image.

Recommendation on the second opinion

Keep **SECOND_OPINION_MODE=shadow**. The scorer, trained on a public benchmark (RCAEval), did not transfer: it disagreed with 8 of 13 correct rule decisions and named the right service 62% of the time. Gate mode would block correct fixes. A scorer trained on our own faults is built and tested on synthetic data only; it has no accuracy figure until a collection run is made.

AWS Academy limits

- Instance types above *large* are denied.
- No IAM role creation; use LabInstanceProfile.
- Session lasts about 4 hours.
- The S3 console serves .jsonl files as downloads.

Still open

- Bucket **autosre-validation-841736140404** and an unused security group remain. The bucket holds the raw run data.
- The VLE*-jenkins instances in the lab account may still be running and using credits; they were not touched.
- Next run: `COLLECT=1 PER_CAUSE=8 UNKNOWN=6 ./scripts/aws/bootstrap.sh <bucket>` collects data to train the scorer on our own faults (about 2.5 hours of lab time).